



Advanced Obfuscation and Diversification

CEN429 Secure Programming — Week 9

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Today's Plan (3 Hours)

Hour	Section	Topic
1	1–2	Why is obfuscation a rule? · MATE · taxonomy · rule template
2	3	Control-flow rules (K-01...K-06)
3	4–6	Data rules · virtualisation/dynamic · diversification · measurement

Learning outcome (LO.3): write obfuscation as a security rule · state each technique's cost and limit · measure a protection and justify it

What We Bring from Earlier Weeks

- **The white-box / MATE attacker model** — the threat model where whoever holds the device is also the attacker (Week 1)
- **Debugger** — a tool that runs a program step by step and shows variables (gdb, lldb); we saw its run-time detection in the RASP context (Week 6)
- **Entropy** — a measure of how disordered a data's bytes look; we saw it in detecting encrypted/packed content and in random number generation (Weeks 2, 3)
- **Compiler flags, strings and the symbol table** — compiler options and the tools that list a binary's readable text/name list; the entry-level steps of obfuscation (Week 4)
- **Decompilation** — turning a binary/bytecode back into something close to readable code; we saw it on JVM bytecode (Week 5)
- **XOR** — 1 if the two bits differ, 0 if they match; $a \oplus b \oplus b = a$ undoes itself; we decoded it by hand in Java string obfuscation (Week 5)
- **Logging** — the informational messages a program writes while running; if sensitive information ends up in a log, the attacker reads it (Week 2)

This week: we use these tools and concepts at an **advanced** level and in a **measurable** way.

This Week's Concepts

Each term is defined once, where it first appears in the body; here we only mark **where**.

Concept	Where
Obfuscation as a security rule, MATE	Section 1
Obfuscation taxonomy (five families)	Section 2
The "protection rule" template (protects what/cost/limit/measurement)	Section 2
Control-flow rules (K-01–K-06)	Section 3
Data obfuscation rules (K-07–K-09)	Section 4
Whole-program rules (K-10–K-12)	Section 4
Diversification	Section 5
Obfuscation's four metrics (potency/resilience/stealth/cost)	Section 5
Deobfuscation (the other side's tools)	Section 5
Its place in layered defence, project S9	Section 6

What Is Source Code?

- The **human-readable** program text that **you** write (e.g., a C file).
- Example: `int topla(int a, int b) { return a + b; }`

Source code is **compiled** and turned into the form the machine runs.

Compiler and Binary

- **Compiler:** the program that translates source code into **machine code** (gcc, clang).
- **Binary:** the result of compilation; the file the computer runs directly (`.exe` , `.so`).

Derleyici ne yapar: kaynaktan ikiliye

gizleme işte bu iki uçtan birinde devreye girer



Saldırganın elinde yalnız sağdaki kutu vardır; tersine mühendislik soldakini geri kurmaya çalışır.

Machine Code and Assembly

- **Machine code:** the numeric instructions the processor understands.
- **Assembly:** a somewhat more human-readable form of machine code (`mov` , `cmp` , `jmp`).

This is what you see when you open a binary.

What Is Reverse Engineering?

Reverse engineering: looking at a binary and trying to work out **what the program does**.

This is the attacker's basic job.

Decompiler

- **Decompiler:** a tool that turns a binary back into a form close to readable code.
- Examples: Ghidra, IDA.

Goal: someone with no access to the source can infer the logic just by looking at the binary.

Bit, Byte, Hexadecimal

- **Bit:** 0 or 1.
- **Byte:** 8 bits.
- **Hexadecimal (hex):** written with a `0x` prefix; `0x2A` = 42.

We will see values like `0x5A` in the code; these are just numbers.

What Is XOR?

- **XOR** ($\wedge$): 1 if the two bits differ, 0 if they are the same.
- Property: $a \wedge b \wedge b == a \rightarrow$ **it undoes itself**.

```
c = a ^ 0x5A; /* şifrele */  
a = c ^ 0x5A; /* geri çöz */
```

It is heavily used in obfuscation because it is reversible.

What Is Entropy (Randomness)?

- **Entropy:** how "random" a piece of data looks.
- Cryptographic keys look **high-entropy** (irregular bytes).

If an attacker sees a high-entropy block in a binary, they say "there might be a key here."

Function, Branch, Condition

- **Function:** a block of code that does one job (`erisim_ver`).
- **Branch:** a fork in the road, like an `if`.
- **Condition:** the expression that decides which way the branch goes.

Basic Block

- **Basic block:** a sequence of instructions that runs straight through, with no branch.
- When an `if` is reached the block ends, and two new blocks begin.

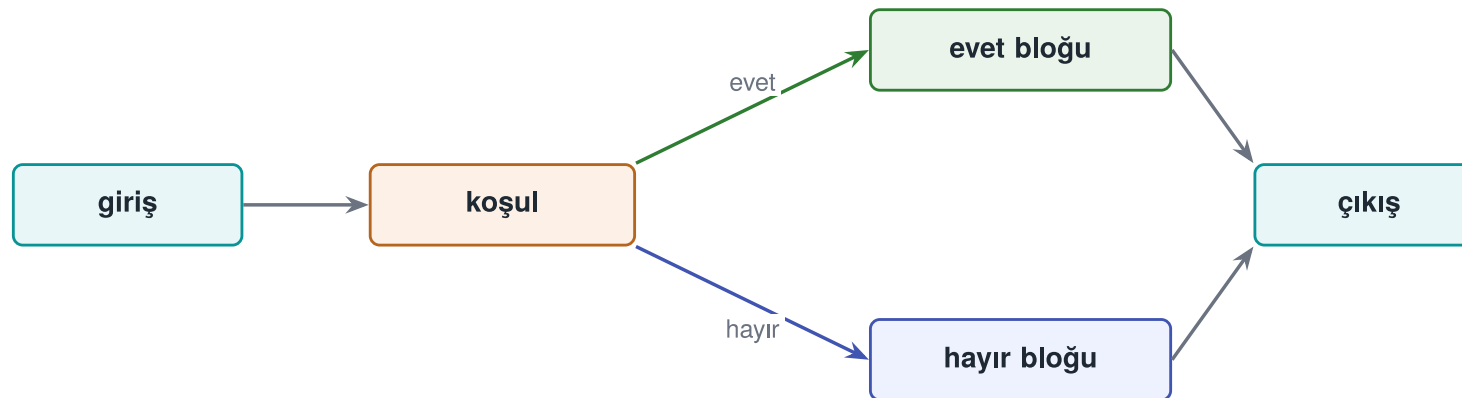
Program = basic blocks wired together.

Control Flow Graph (CFG)

- **CFG** (Control Flow Graph): a diagram that makes basic blocks its **nodes** and the transitions between them its **edges**.
- It is the program's "road map."

Kontrol akışı grafiği (CFG)

tersine mühendisliğin ilk çıkardığı iskelet



Düzleştirme bu iskeleti tek bir dağıtıcıya indirir: “hangi blok ne zaman” bilgisi kaybolur.

This Week's Question in One Sentence

I have delivered the program to the user; **the attacker now owns it**. How do I make it **harder** for them to read and modify my code?

Answer: **code obfuscation rules**. Let's begin.



1. Why Is Obfuscation a Rule?

First, a Question

You have **delivered** a mobile app to the user.

- Whose machine is the code running on now?
- Are your server-side security checks still valid here?
- How many times can the attacker try?

MATE: the Attacker at the End

Man-At-The-End = an attacker who possesses the program itself.

- The binary is in their hands
- They can read and modify memory
- They have a decompiler and a debugger in hand
- They can try as many times as they like

This is also called the **white-box** model (week 1).

The MATE Attacker Model — Diagram

MATE: saldırgan cihazın sahibi

mobil ödeme, DRM, lisanslama: kod kullanıcının elinde çalışır

Ağ saldırganı

yalnız mesajları görür
TLS onu durdurur

klasik güvenlik dersleri
bunu varsayar

MATE (uçtaki insan)

ikiliyi açar, belleği okur
debugger bağlar, yamalar
istediği kadar dener

kripto TEK BAŞINA yetmez

Gizleme burada devreye girer: kırılmazlık değil, MALİYET yükseltme.

Black Box ≠ White Box

	Black box	White box (MATE)
Sees	Input/output	Everything
Environment	Remote server	The attacker's device
Attempts	Limited	Unlimited
Defence	Crypto, verification	+ obfuscation, RASP, whitebox

Cryptography assumes a **black box**; that assumption collapses on delivery.

A Brief History — Where Does Code Obfuscation Come From?

- **1976** — Diffie & Hellman: the idea of "incomprehensible but working" (protection **through effort**)
- **1997** — Collberg et al. **taxonomy** + *potency–resilience–stealth–cost* (this course's five families)
- **2001** — Barak et al.: perfect obfuscation is **impossible** → not "unbreakability" but **cost**
- **2002** — Chow et al. **whitebox AES** (week 11) · **2010s Tigress**, O-LLVM (week 14)

Obfuscation is a **practical delaying** discipline built on top of an **impossibility theorem**.

The Honest Truth

An attacker with enough time, skill, and motivation breaks **every obfuscation**.

Cookbook (Recipe 12.1): anti-tampering is not "unbreakable."

So what is the goal, then?

The Goal: Make Breaking It Uneconomical

Obfuscation is a tool for **raising cost**, not for keeping a secret.

- Make breaking it **expensive**
- Make automating it **harder**
- **Buy time** for the other layers

Let's turn these into three rules.

Rule 1 — Push the Cost Above the Value

The attack must be **more expensive** than the protected asset.

- Protecting a 10 TL secret with 1000 TL of effort: a waste
- Protecting a 1000 TL secret with 10 TL of effort: insufficient

First determine the asset's value, then choose the protection.

Rule 2 — Break Automation

An attack that works on one copy should **not** work on **every** copy.

- This is called **diversification** (section 5 of this week)
- With tooling in week 14 (Tigress --Seed)

One crack should not open every door.

Rule 3 — Buy Time for the Other Layers

Obfuscation is **not** a defence on its own; it is a **layer**.

- RASP (week 6) protects it while it is running
- Integrity checking, server-side checks
- Holding out **until** the key/version is renewed is enough

The Source's Language: Every Countermeasure Is a "Rule"

The technical guide writes every countermeasure under three headings:

- **Description** — what it protects
- **Implementation** — how it is done
- **Example usage** — where

Usually with a **compliance ID** too (e.g., "sensitive data is never logged in plain text").

We will write it the same way: *what it protects · how · cost · limit*.



2. Taxonomy and Rule Template

What Does Obfuscation Hide?

Collberg's classification: we split obfuscation by **what it hides**.

There are five families. Let's go through them in order.

Family 1 — Layout

What does it hide? Names, format, metadata.

- Making symbols invisible
- Making function/file names meaningless
- Stripping logging from the release build

We met this in week 4; this week we go deeper.

Family 2 — Data

What does it hide? Constants, strings, variables.

- String encoding
- Constant transforms
- Variable splitting/merging
- Opaque boolean

→ Section 4 of this week.

Family 3 — Control Flow

What does it hide? The algorithm's structure.

- Control-flow flattening
- Opaque predicates
- Bogus and dead branches
- Random exit

→ Section 3 of this week (coming up shortly).

Family 4 — Anti-Analysis

What does it hide? The analysis tools' own work.

- Structures that mislead a decompiler
- Decoding code at runtime

→ Concept this week, RASP in week 6.

Family 5 — Virtualisation

What does it hide? The machine code itself.

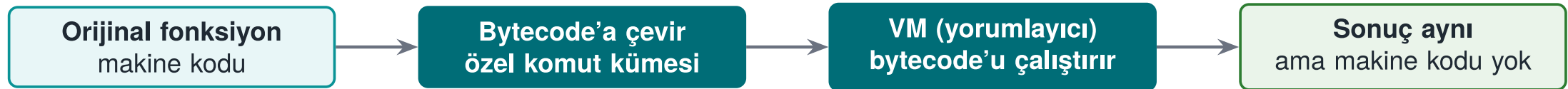
- Turning a function into a custom VM's bytecode

→ Concept this week (K-10), with tooling in week 14 (Tigress `Virtualize`).

Virtualisation — Diagram

Sanallaştırma: en güçlü, en pahalı

analizci artık sizin uydurduğunuz makineyi çözmek zorunda



Maliyet: onlarca kat yavaşlama + boyut → yalnız küçük ve kritik fonksiyonlara.

The Five Families: Summary

Family	Hides
Layout	Name, format
Data	Constant, string, variable
Control flow	Structure
Anti-analysis	Tool's work
Virtualisation	Machine code

Rule: the families are used **together**, not alone.

Five Families — Potency ↑ Cost ↑

Gizlemenin beş ailesi

aşağıdan yukarı: güç ↑ · maliyet ↑



Güç birliktelikten ve çeşitlendirmeden gelir; her aile ölçülmeli. Hiçbiri anahtarı korumaz.

In the Field: All Together

The native countermeasures listed in a certified product cover this **entire** map:

symbol hiding · name obfuscation · arithmetic transform · string encoding · opaque boolean · bogus operation · dead branch · flattening · random exit · stripping logging

None of them is strong alone; strength comes from **togetherness**.

What Obfuscation GIVES

- A **delay** against static analysis (`strings` , symbols, decompilation)
- **Resistance** against automated attack (via diversification)
- **Time** for the other layers
- A **measurable** increase in cost

What Obfuscation DOES NOT Give

- The **secrecy** of a secret → that is cryptography's job
- **Protection** against reading memory → RASP, whitebox
- **Permanent** security → there is no such thing as "unbreakable"
- **Free** protection → every obfuscation comes with a cost

What Obfuscation Gives and Does Not Give — Diagram

Gizleme ne verir, ne vermez?

gizleme bir maliyet kuralıdır, bir güvence değil

VERMEZ

gizlilik (kriptonun işi)
matematiksel güvence
kalıcı koruma
kötü tasarımı kurtarma

VERİR

saldırgana ZAMAN kaybettirir
otomatik araçları zorlaştırır
toplu kırılmayı engeller
tespit için süre kazandırır

“Ne kadar sürer?” sorusuna cevap verir; “kırılamaz mı?” sorusuna değil.

The Most Common Mistake

Mistaking obfuscation for **storing a key**.

- An embedded/hidden key is still **there**
- The program has to use it while it runs → it is exposed in memory
- The decoding logic is inside too

Secrecy is cryptography's job, not obfuscation's. (week 3)

The "Protection Rule" Template

```
KURAL K-xx: <teknik>
  Neyi korur?   : hangi varlık / kod bölümü
  Hangi tehdit?: statik / dinamik / otomatik / kurcalama
  Nasıl?        : bir cümle (kavram)
  Maliyet       : boyut, hız, bakım, hata ayıklama
  Sınır         : neyi korumaz
  Ölçüm         : etkinlik nasıl doğrulanır
```

Why Is the "Measurement" Line Required?

Obfuscation is not "present/absent."

- The evaluator (week 13) asks **how much each layer delays** the attack
- Protection without measurement = a **claim**, not evidence

In the project you will write S9 using this template.

Four Metrics — Diagram

Gizlemeyi ölçen dört boyut

bir koruma kararı bu dördüyle birlikte gerekçelendirilir

GÜÇ (potency) — insan analizciyi ne kadar zorlaştırır

DAYANIKLILIK (resilience) — otomatik araçlara direnç

GİZLİLİK (stealth) — gizlemenin fark edilmezliği

MALİYET (cost) — boyut + hız cezası

Ödünleşim: dayanıklılık ↔ maliyet, güç ↔ gizlilik. Ölçmeden “güçlendirdim” denmez.

Section 2 — Quick Check

1. How does the MATE attacker differ from a black box?
2. What were obfuscation's three rules?
3. Does obfuscation protect a key? Why?

Section 2 — Answers

1. **MATE** possesses the program (the code/memory is theirs, unlimited attempts); a black box only sees input/output.
2. Push the cost above the value · break automation (diversify) · buy time for the other layers.
3. **No**. The key is exposed in memory while running and the decoding logic is inside; secrecy is cryptography's job.



3. Control-Flow Rules

The Purpose of This Section

We treat the CFG (already defined in the background section) as this section's skeleton; the attacker's first job is to extract that skeleton.

In week 4 we **introduced** flattening.

In this section we add the rules that **strengthen** it:

K-01 opaque predicate · K-02 arithmetic encoding · K-03 bogus/dead · K-04 flattening (in depth) ·
K-05 random exit · K-06 call hiding

All examples are **synthetic**, built around a single check: `erisim_ver`.

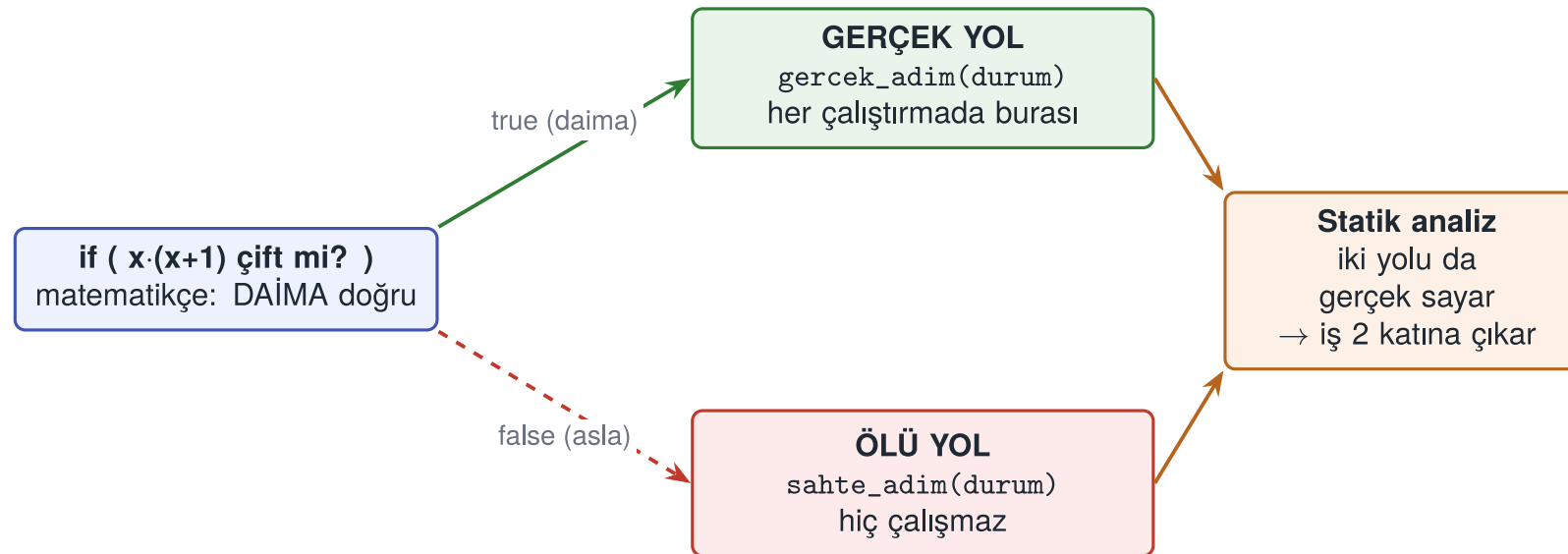


K-01 · Opaque Predicates and Loops

Opaque Predicate — Diagram

Opak yüklem: gerçek yol ile ölü yol

program hep aynı çalışır, analizci iki yolu da incelemek zorunda kalır



Maliyet düşük · Sınır: klasik kalıplar araçlarca tanınır → tohumla çeşitlendirin (14. hafta).

K-01 · What Does It Protect?

- **Which condition** a branch is taken under
- **How many times** a loop runs

Idea: branch on an expression whose value the programmer knows but the tool cannot easily resolve.

K-01 · Opaque Predicate Example

```
/* x*(x+1) daima çifttir → sonuç daima true */  
static int opak_dogru(unsigned x) {  
    return ((x * (x + 1)) & 1u) == 0;  
}
```

The programmer knows: it is **always true**. Static analysis cannot easily prove this.

K-01 · How Is It Used?

```
if (opak_dogru(sayac)) {  
    durum = gercek_adim(durum);    /* daima buraya girer */  
} else {  
    durum = sahte_adim(durum);    /* ölü yol: hiç çalışmaz */  
}
```

The analyst thinks both paths are "real"; they spend their time on the fake path.

K-01 · Opaque Loop

The guide makes loops a separate rule:

- for/while are **never shown directly**
- Hidden **inside** flattening
- The body is split into small blocks
- Iteration is driven by a **state variable**

The information "this loop runs N times" cannot be read off the CFG.

K-01 · Cost and Limit

- **Cost:** low–medium (extra arithmetic/branch)
- **Limit:** well-known opaque predicate patterns are recognised automatically
- **Countermeasure:** diversify (section 5)
- **Measurement:** basic block count before/after



K-02 · Arithmetic Encoding

K-02 · What Does It Protect?

Constants and simple computations.

It hides **directly readable** hints like "this number is 0x1F, this operation is an XOR."

K-02 · Equivalent Expressions (MBA)

Replace an operation with an equivalent one that produces the same result but looks complex:

```
a + b    ≡    (a ^ b) + 2*(a & b)
x * 2    ≡    x << 1
```

Same result, complex appearance. (MBA = mixed boolean-arithmetic)

K-02 · Hide the Constant

```
/* 0x2A doğrudan yazılmaz; iki parçadan üretilir */  
static uint8_t esik(void) {  
    uint8_t a = 0x37, b = 0x1D;  
    return (uint8_t)(a ^ b);    /* = 0x2A */  
}
```

Searching the binary for `0x2A` returns nothing.

K-02 · Cost and Limit

- **Cost:** low
- **Limit:** MBA simplifiers (arybo/msynth) can undo many expressions
- **Rule:** weak on its own → **combine** it with control flow
- **Measurement:** a byte/ strings search for the constant should return nothing



K-03 · Bogus Operations and Dead Branches

Bogus Operation $\neq$ Dead Branch — Diagram

Sahte işlem $\neq$ ölü dal

ikisi de sahte iş üretir ama farklı biçimde

SAHTE İŞLEM (bogus op)
ÇALIŞIR
sonucu kullanılmaz
analizciyi ve aracı yorar
maliyet: hafif yavaşlama

ÖLÜ DAL (dead branch)
HİÇ ÇALIŞMAZ
opak yüklemle korunur
analizde gerçek görünür
maliyet: boyut

Sınav sorusu: hangisi çalışır? → Sahte işlem çalışır, ölü dal çalışmaz.

K-03 · The Two Are Different!

The guide makes these **separate** rules. The difference is critical:

- **Bogus operation:** runs, does not change the result
- **Dead branch:** never runs (protected by an opaque predicate)

K-03 · Comparison

	Bogus operation	Dead branch
Does it run?	Yes	No
Result	Unchanged	(does not run)
Purpose	Hide the real thing in a crowd	Show a fake path

K-03 · Bogus Operation Example

```
crc = crc32_guncelle(crc, veri, n);  
crc ^= (sabit ^ sabit);    /* = crc; sonuç aynı */
```

`sabit ^ sabit == 0` → the CRC does not change, but the code gets more crowded.

K-03 · Dead Branch Example

```
if (opak_yanlis()) {           /* daima false */  
    sahte_kontrol();          /* hiç çalışmaz */  
}
```

The analyst treats this block as "real" and examines it; they waste time.

K-03 · Cost and Limit

- **Cost:** low (bogus operation) → medium (dead branches grow the CFG)
- **Limit:** dead code elimination drops the branch once the opaque predicate protecting it is solved
- **Dependency:** depends on the opaque predicate's **quality**
- **Measurement:** time needed to tell dead branches apart from real ones



K-04 · Control-Flow Flattening

K-04 · What Does It Protect?

The algorithm's **structure**:

- The natural adjacency of blocks
- The information "which block leads to which"

K-04 · Idea: Vertical → Horizontal

- **Normal:** blocks follow one another (vertical flow)
- **Flattened:** all blocks sit in a single `switch` dispatcher (horizontal)
- The order can only be read from a **state variable**

K-04 · Template

```
int durum = BASLA;
for (;;) switch (durum) {
    case BASLA: durum = ADIM1; break;
    case ADIM1: durum = kosul ? ADIM2 : HATA; break;
    case ADIM2: durum = BITIR; break;
    case HATA: return RED;
    case BITIR: return IZIN;
    default: return RED; /* rastgele çıkış (K-05) */
}
```

K-04 · Weak on Its Own

We said back in week 4: an experienced analyst can solve this template fairly quickly.

Strength comes from reinforcements. In order:

K-04 · Reinforcement 1 — State Values (K-02)

- Scattered, arithmetically produced values instead of case 1, 2, 3
- The sequence is **invisible**
- Which step follows which cannot be read

K-04 · Reinforcement 2 — Bogus/Dead (K-03)

- Bogus case s are added
- Telling which case is **real** becomes harder

K-04 · Reinforcement 3 — Random Exit (K-05)

- Errors exit through the `default` branch with an **unpredictable** state
- **Where** the failure happened cannot be read

K-04 · Reinforcement 4 — Opaque Transitions (K-01)

- `case` transitions depend on **opaque conditions**, not fixed values
- Static tracing becomes even harder

K-04 · Cost and Limit

- **Cost:** medium–high (block/branch count grows a lot, performance drops)
- **Limit:** flattening alone → can be undone with symbolic execution (section 5)
- **Strength:** comes from the reinforcements
- **Measurement:** CFG node/edge count, tracing time



K-05 · Random Exit

K-05 · What Does It Protect?

Where a check fails.

The attacker typically:

- Finds the "failure branch"
- Turns it into the "success branch"

K-05 · How?

```
if (!imza_gecerli(p)) {  
    durum = kararsiz_deger(); /* switch'te tanımsız → default */  
    break;  
}
```

There is no direct `return RED`; the state goes to an unpredictable value.

K-05 · Success Should Exit the Same Way Too

- A successful exit uses the same method too
- So success and failure **look alike** in the flow
- The analyst cannot say "this is the accept branch"

K-05 · Cost and Limit

- **Cost:** low
- **Limit:** not meaningful on its own, only together with flattening
- **Goal:** a single-byte patch should **not be enough** to bypass the check
- **Measurement:** time needed for a "find and flip the failure branch" attack



K-06 · Call and Dependency Hiding

K-06 · What Does It Protect?

Hints like: "this function calls `memcmp` , so it must be doing a comparison."

Standard library calls **tell** the attacker what you are doing.

K-06 · How?

- Your **own** constant-time `esit_mi` instead of `memcmp` on the critical path
- No recognisable name **appears** to the linker
- **Bogus parameters:** add parameters that are never used → the signature becomes misleading

K-06 · Do Not Break Constant-Timeness

When you write your own comparison, make sure it stays **constant-time** (week 3).

- An early-exit comparison → opens a **side channel**
- Do not create a **new hole** while adding a protection

K-06 · Cost and Limit

- **Cost:** low–medium (your own versions need maintenance)
- **Limit:** behavioural analysis can still reveal what it does → this is a **delay**
- **Measurement:** count of recognisable library calls in the binary

Section 3 — Summary

- K-01 opaque predicate/loop · K-02 arithmetic encoding
- K-03 bogus operation $\neq$ dead branch
- K-04 flattening (with reinforcements)
- K-05 random exit · K-06 call hiding

None is sufficient on its own; **together** and **measured**.

Section 3 — Quick Check

1. What is the difference between a bogus operation and a dead branch?
2. What are the 4 rules that strengthen flattening?
3. Which attack does random exit make harder?

Section 3 — Answers

1. A **bogus operation** runs but does not change the result; a **dead branch**, protected by an opaque predicate, never runs at all.
2. Hide the state values (K-02) · bogus/dead blocks (K-03) · random exit (K-05) · opaque predicate transitions (K-01).
3. The "find the failure branch and flip it to success" attack; a **single-byte patch is no longer enough**.



4. Data and Whole-Program Rules

Data Obfuscation Rules K-07–K-09 — Diagram

Veri gizleme kuralları (K-07 – K-09)

amaç: statik okumada anlamlı veri bırakmamak

K-07 Dize kodlama — sabit metinler şifreli saklanır, kullanım anında çözülür

K-08 Opak boolean — karar değeri hesaplanır, sabit karşılaştırma yapılmaz

K-09 Değişken bölme — bir değer birden çok parçada tutulur, geç birleştirilir

Ortak tuzak: çözülen değeri gereğinden uzun süre bellekte tutmak — kazancı siler.

Control Flow $\neq$ Data

- **Control flow** (section 3): hides the algorithm's **structure**
- **Data obfuscation** (this section): hides the **values** the program processes

Strings, constants, tables, variables.



K-07 · String Encoding

K-07 · What Does It Protect?

The **readable text** inside a binary.

The cheapest attack step: `strings` .

`"Lisans gecersiz"` or a URL → tells the attacker exactly where to look.

K-07 · Idea

- Sensitive strings are encoded **before compilation** (e.g., XOR)
- They stay **encoded** in the binary
- Decoded **at the moment of use**
- **Wiped** the instant the job is done

K-07 · Example (Synthetic)

```
static const uint8_t GIZLI[] = {0x3B,0x2A,0x2E,0x2E,0x2D};
void kullan(void) {
    char tmp[sizeof GIZLI];
    for (size_t i=0;i<sizeof GIZLI;i++)
        tmp[i] = GIZLI[i] ^ 0x5A;      /* çöz */
    isle(tmp, sizeof GIZLI);
    memset_s_benzeri(tmp, sizeof tmp); /* hemen sil */
}
```

K-07 · Critical Limit

- The decoded string is **exposed in memory** while running
- The decoding key is also in the binary
- This is a **static-scan** countermeasure

In the field: the decoding key is **split and distributed**, and the decoding function gets extra checks.

K-07 · String Obfuscation ≠ Key Storage

Again: string obfuscation stops `strings` , it does **not** protect a **key**.

- For the key → whitebox (week 11) or hardware
- **Measurement:** the sensitive string **must not** be found in `strings` output



K-08 · Opaque Boolean

K-08 · Problem: Plain 0/1 Is Dangerous

Returning a check's result as a plain 0 / 1 :

- The attacker flips a **single byte**
- "Failure" becomes "success"

K-08 · Idea

Tie the true/false value to:

- **Multiple values**
- In a way a single-byte patch cannot flip

Functions return an **opaque return code** instead of a plain boolean.

K-08 · Example (Concept)

```
typedef struct { uint32_t a, b; } Karar;  
static Karar izin_ver(void) {  
    return (Karar){ 0xA3C1u, 0x5C3Eu }; /* a ^ b == 0xFFFF */  
}  
static int karar_izin_mi(Karar k) {  
    return (k.a ^ k.b) == 0xFFFFu;  
}
```

The calling side verifies **both fields**.

K-08 · Cost and Limit

- **Cost:** low
- **Limit:** solvable with enough scrutiny
- **Goal:** block a single-byte patch and crude branch-flipping
- **Measurement:** independent points > 1 needed to flip the result



K-09 · Variable Splitting and Restructuring

K-09 · What Does It Protect?

The **recognisable footprint** of a sensitive variable in memory.

Patterns like "a 32-byte block = an AES key" point the attacker straight at it.

K-09 · How?

- **Split** a variable (produce a 32-bit value from two 16-bit shares)
- **Merge** several variables into a single word
- Rearrange arrays
- Change buffer size/order/layout
- **Fill with random values** after use

K-09 · Cost and Limit

- **Cost:** low–medium
- **Limit:** only slows down static/pattern analysis
- **Measurement:** time needed to recognise the sensitive buffer from the pattern



Whole-Program Rules: K-10, K-11, K-12

K-10 · Virtualisation (Concept)

What does it protect? A function's **machine code**.

How? The function → a custom **VM's bytecode**; the bytecode and a small interpreter are embedded in the binary.

The attacker does not see familiar code — they see a **custom instruction set** they must first solve.

K-10 · Cost and Limit

- **Cost: high** (interpreted code is slow; maintenance is expensive)
- **Use:** only the most critical, small, rarely-changing functions
- **Limit:** once the VM is solved once, everything it protects opens up; if not diversified, every copy is identical
- Week 14: Tigress `Virtualize`

K-11 · Compiler-Based (O-LLVM)

What does it protect? The rules above, applied through a **compiler pass**, not by hand.

- LLVM-based obfuscators: flattening, bogus control flow, instruction-substitution passes
- The source stays **readable**; protection is a compile option
- A different **seed** per build → easy diversification

K-11 · Cost and Limit

- **Cost:** medium (depends on the passes selected)
- **Limit:** known pass patterns are recognisable; stay current with the tool's version
- **Measurement:** the difference between binaries produced with different seeds

K-12 · Dynamic Encryption (Concept)

What does it protect? The most sensitive code sections, by keeping them **encrypted** in the binary.

How? The section stays encrypted → decoded only right when it is about to run → then re-encrypted/wiped.

K-12 · Risk

- Directly **conflicts** with OS memory protections (**DEP/NX, W^X**)
- Done wrong, it both crashes and opens a **new hole**
- The code sits exposed in memory for a moment while running → a memory dump can catch it

K-12 · Rule

If adding one protection **turns off** another, measure the net gain and record the trade-off.

In practice it is used very selectively, on small sections.

The Java/Managed Side (from Week 5)

- This week's rules are for **native** (C/C++)
- The managed-language equivalent: **ProGuard/R8** name obfuscation + dead code elimination + shrinking
- APIs reached via `-keep` and reflection

The Two Sides Work Together

- The Java side is obfuscated with R8
- The native side, with this week's rules
- **Cross-check:** native verifies Java's integrity, Java verifies native's (week 6, RASP)

Section 4 — Summary

- **K-07** string encoding · **K-08** opaque boolean · **K-09** variable splitting
- **K-10** virtualisation (expensive) · **K-11** compiler-based (O-LLVM) · **K-12** dynamic (risky)

Data obfuscation **complements** control-flow obfuscation.

Section 4 — Quick Check

1. Does string obfuscation protect the key?
2. Why does an opaque boolean make a single-byte patch harder?
3. What are virtualisation's two biggest costs?

Section 4 — Answers

1. **No**. The decoded string is exposed in memory; the decoding key is inside; it only stops **static** (`strings`) scanning.
2. The result depends on multiple values; flipping it requires changing **>1 independent points**.
3. High **performance** cost (tens of times slower) and high **maintenance** cost; that is why it is used only on small/critical functions.

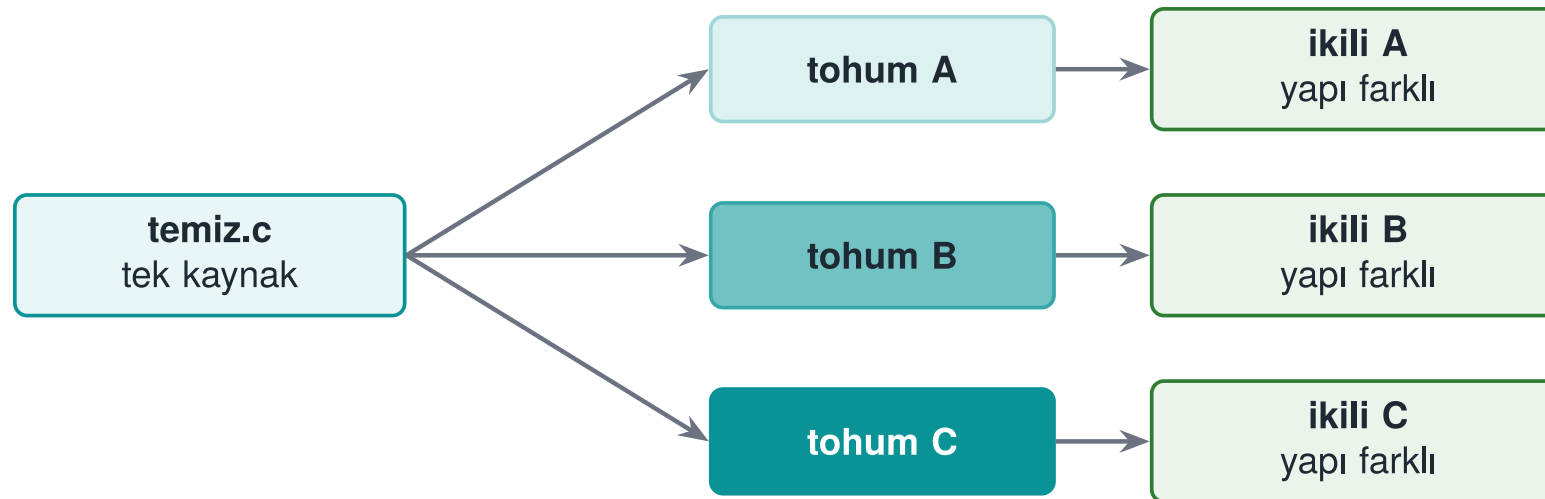


5. Diversification and Measurement

Diversification — Diagram

Çeşitlendirme: tek kırık her yeri açmasın

aynı kaynak, farklı tohum, farklı ikili



Davranış birebir aynı; bir kopyaya yazılan yama diğerlerinde ÇALIŞMAZ.

Diversification: Recall Rule 2

An attack that works on one copy should not work on every copy.

If the attacker can crack one copy and distribute the patch/script to **everyone**, one crack opens every door.

What Is Diversification?

Producing binaries from the same source that are **behaviourally equivalent, structurally different**.

Opaque predicates, bogus blocks, state values **change** from copy to copy.

Two Kinds of Diversification

- **In space:** every build/distribution gets a different **seed**
 - An attack written for one copy does not work on another
- **In time:** every release gets a new arrangement
 - An attack written for the old release breaks in the new one

Works together with key/version renewal (week 10).

What Does Diversification Prevent?

- It does not raise obfuscation's **potency** (a single copy can still be cracked)
- But it prevents **scaling**

Week 14: with tooling, via Tigress --Seed , RandomFuns .

Diversification Effectiveness: a Simple Metric

- Compile the same source with **two different seeds**
- Measure the **byte difference** of the protected functions
- The higher the difference, the lower the odds that an attack on one works on the other

Write this metric into S9.



Measuring Obfuscation

Why Measure?

To **defend** a protection decision.

Collberg's framework: four dimensions.

These four are the foundation of week 13's **attack potential** scoring.

Metric 1 — Potency

Question: how incomprehensible is it to a human?

How: complexity metrics — CFG node/edge count, cyclomatic complexity, nesting depth (before/after).

Metric 2 — Resilience

Question: how much does it withstand an automated tool?

How: how long/successfully a deobfuscation tool (symbolic execution, a simplifier) takes to **undo** the protection.

Metric 3 — Stealth

Question: does the obfuscation give itself away?

How: is obfuscated code statistically distinguishable from normal code?

If it is distinguishable, the attacker knows where to look.

Metric 4 — Cost

Question: how much does it cost us?

How: size increase, runtime, memory, maintenance, debugging difficulty.

Before Flattening (Concept)

Kontrol akışı düzleştirme: CFG önce ve sonra

aynı program, aynı sonuç — farklı iskelet

ÖNCE: okunur akış
giriş → koşul → dallar
döngü yapısı görünür
analizci mantığı
doğrudan okur

SONRA: düzleştirilmiş
tek switch (dağıtıcı)
bloklar durum değişkeniyle
sırası gizli
yapı bilgisi KAYBOLUR

Tek başına zayıf: dağıtıcı deseni tanınır → opak yüklem + durum şifreleme ekleyin.

Few nodes, readable flow.

After Flattening (Concept)

Many nodes, a single hub; "which block comes next" cannot be read.

Potency Measurement Example

Metric	Before	After
Basic blocks	5	34
Edges	6	60
Cyclomatic complexity	3	28

You take these numbers from the decompiler and write them into S9.

Cost Measurement Example

```
size ./program_temiz ./program_gizli # boyut  
# süre: aynı girdiyle N kez çalıştır, ortalama
```

Metric	Before	After
Size	100 KB	128 KB
Runtime	1.0×	1.7×

Interpreting the Measurement

- Potency ↑ (34 blocks) and cost ↑ (28% size, $1.7\times$ time)
- This **trade-off** is expected
- Decision: is the gain acceptable given the asset's value?

Do not say "strong"; show **these numbers**.

The Four Metrics Trade Off

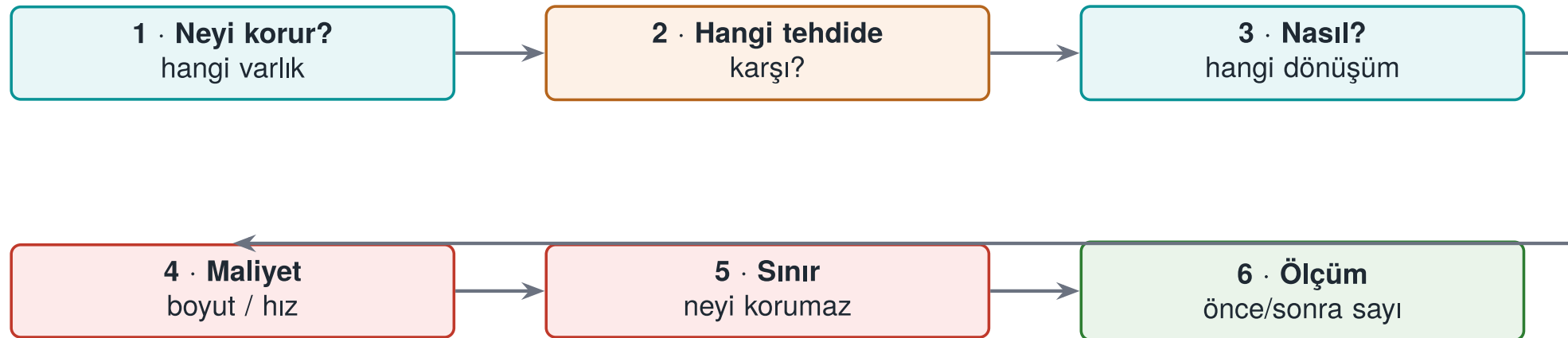
Raising	Result
Potency ↑ / resilience ↑	Cost ↑
Complexity ↑	Stealth ↓ (draws more attention)

Good decision: balance the four against the **value of the asset**.

Decision Rule (Write into S9)

“Koruma kuralı” şablonu

her teknik bu altı satırla yazılır — S9’a böyle gider



“Ölçüm” satırı olmayan kural, gerekçelendirilmemiş sayılır.



Deobfuscation and Resilience

Why Should We Know the Other Side's Tools?

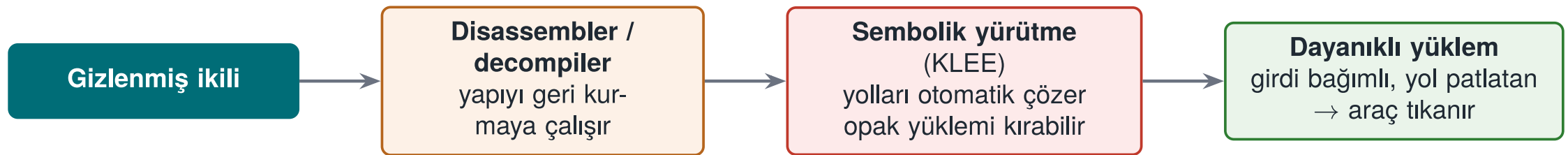
To measure resilience correctly.

We learn these **to test our own defence** — not to attack.

The Other Side's Tools — Diagram

Karşı tarafın araçları

dayanıklılık, bu araçlara karşı ölçülür



Klasik kalıplar kütüphanelerde tanınır → tohumla çeşitlendirin (14. hafta).

Tool 1 — Symbolic Execution (KLEE)

- Program paths are solved as mathematical constraints
- Opaque predicates and flattening can be undone this way

Resilience rule: tie opaque predicates to structures that are expensive to solve (factorization, hashing), grow the state space — but **measure the cost**.

Tool 2 — MBA Simplifier

- Undoes arithmetic encoding (K-02)
- Countermeasure: also tie encoded constants to opaque predicates — layers that depend on each other are harder to solve separately

Rule: do not **count** arithmetic encoding as a secrecy layer on its own; combine it with control flow.

Tool 3 — Pattern Recognition

- Recognises known library/pass patterns

Rule: diversify; do not rely on a single opaque predicate pattern or a single VM design.

(in space: several patterns/seeds · in time: change the patterns every release)

Main Takeaway

Banescu et al. (Tigress + KLEE): how much each transform withstands symbolic execution is **measured**.

Resilience is not a **claim**, it is a **measured** quantity.

Do not say "strong"; **produce a number**: "this transform pipeline did not let this problem, of this size, be solved in this much time, against this tool."

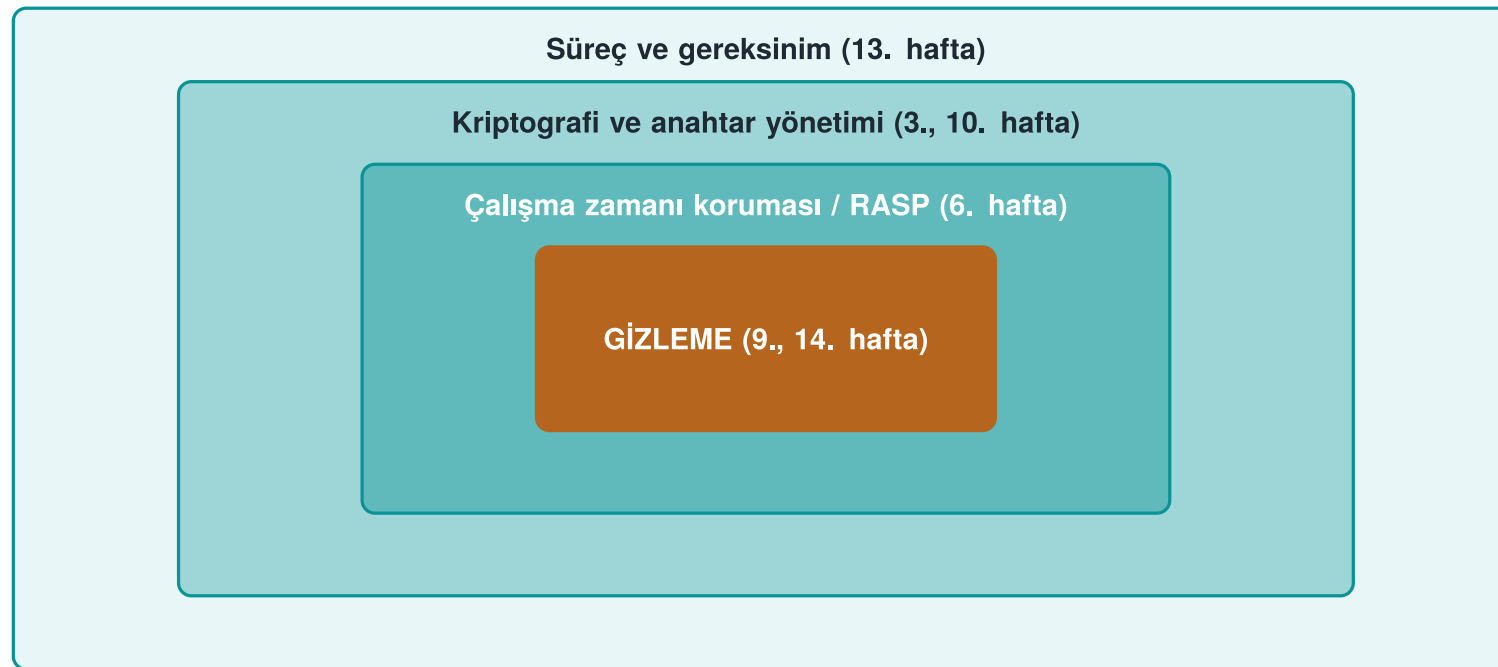


6. Layered Defence, Project, Closing

Obfuscation in Layered Defence — Diagram

Katmanlı savunmada gizlemenin yeri

gizleme en dıştaki kabuk değil, diğerlerini örten bir katmandır



Tek başına gizleme bir güvenlik mimarisi değildir; diğer katmanların ömrünü uzatır.

Obfuscation's Place in Layered Defence

Layer	Week	Relationship
Memory safety/secure compilation	4	Does not replace it; works together
RASP	6	Obfuscation also hides RASP; RASP protects obfuscation while it runs
Whitebox	11	The real protection for the key; obfuscation is the shell
Key renewal	10	Time "until it is renewed"
Certification	12–13	The delay is scored

An Observation from the Field

In a certified product, the native countermeasures cover **all** of this week's rules.

But the document notes, next to every one of them: "**not strong on its own.**"

What convinces an evaluator: togetherness + diversification + **measured** delay.



End to End: Layer by Layer Protection

The Goal of This Example

We take a single synthetic check (`erisim_ver`) and protect it **layer by layer**.

At every step: what was added · what we gained · what we paid.

Step 0 · Unprotected Start

```
int erisim_ver(const char *jeton) {  
    if (jeton_gecerli(jeton))  
        return IZIN;      /* tek dal, tek dönüş */  
    return RED;  
}
```

Weakness: a single branch, `strings`, bypassed with a single-byte patch.

Step 0 · What Does the Attacker Do?

- `strings` → `IZIN` / `RED` and the related strings
- Decompile → a single `if`, two returns
- Flip the failure branch to success (a single byte)

Time: minutes. **Skill:** low.

Step 1 · K-07 Encode the String/Constant

```
/* IZIN/RED ve dizgeler artık düz görünmez */  
static int izin_kodu(void){ return 0x9E ^ 0x9F; } /* =1 */
```

- **Gain:** strings does not find the sensitive text
- **Cost:** very low

Step 1 · Measurement

Metric	Step 0	Step 1
Sensitive strings in <code>strings</code>	3	0
Binary size	baseline	+0.5%

The first layer is the cheapest and gives the highest return.

Step 2 · K-02 Arithmetic Encoding

```
/* karşılaştırma karmaşık ama denk bir ifadeye döner */  
int esik = ((a ^ b) + 2*(a & b)); /* = a + b */
```

- **Gain:** the constant/computation is not directly readable
- **Cost:** low

Step 3 · K-04 Flatten

```
int durum = BASLA;
for(;;) switch(durum){
    case BASLA: durum = kontrol()?ADIM:HATA; break;
    case ADIM:  durum = BITIR; break;
    case HATA:  return RED;
    case BITIR: return IZIN;
}
```

No single `if` anymore; the order lives in the state variable.

Step 3 · What Changed?

- The CFG is now "horizontal" — the blocks sit in one switch
- Which branch means "success" cannot be read directly
- **Cost:** medium (more blocks/branches, a bit slower)

Step 4 · K-05 Random Exit

```
if (!kontrol()) { durum = kararsiz_deger(); break; } /* →default */
```

- **Where** the error was caught cannot be read from the flow
- A single-byte patch is no longer enough

Step 5 · K-03 Bogus + Dead

```
crc ^= (s ^ s);          /* sahte işlem: sonuç aynı */  
if (opak_yanlis()) sahte(); /* ölü dal: çalışmaz */
```

- The real logic sits in a crowd
- **Cost:** medium

Step 6 · K-08 Opaque Boolean

```
Karar k = izin_ver();           /* a^b==0xFFFF */  
if (karar_izin_mi(k)) uygula();
```

- The result is not a plain 0/1 → a single byte cannot flip it
- Flipping it requires **multiple** points

The Cumulative Effect of the Layers

Step	Rule	Attack time (relative)	Cost
0	—	1×	—
1	K-07	2×	very low
2	K-02	3×	low
3	K-04	8×	medium
4–6	K-05/03/08	20×+	medium

(Relative, for illustration; real values are measured.)

Step 7 · K-10 Virtualisation?

- Only for the **most critical** function
- **Cost:** high (tens of times slower)
- **Unnecessary** for most checks

Rule: do not go this far if the value is low.

End to End: Takeaway

- Every layer made the attack a little more **expensive**
- Every layer added a **cost**
- Where you stop depends on the **asset's value** and on what you **measured**

This table is exactly the protection table you will put into S9.

Project · S9 (Advanced Hardening) — 1

1. Protection table

For 2–3 critical sections (license, key derivation, integrity), fill in the rule template:

what it protects · threat · how · cost · limit · measurement

Project · S9 — 2

2. Measurement

For at least one technique, before/after:

- Count of sensitive strings in `strings` output
- CFG node count
- Binary size, the duration of an operation

Project · S9 — 3, 4

3. **Diversification decision:** yes/no + **justification**

4. **Limit and remaining risk:** what does each protection **not** protect?

"This string obfuscation does not protect the key; for the key see S8 whitebox/hardware."

Later: S15 (week 14, putting obfuscation into the build pipeline) · S16 (week 12, testing that behaviour is unchanged).

Through an Evaluator's Eyes

What S9 is graded on is **not** "I used a lot of techniques."

It is graded on: **the justification and measurement of every technique.**

An unmeasured protection = a claim. Not evidence.

Self-Check (1–6)

1. What is MATE? Why does it leave cryptography insufficient on its own?
2. "Makes it expensive" — give an asset-value example?
3. Bogus operation $\neq$ dead branch?
4. Why is flattening weak on its own? Name 3 reinforcements?
5. Which attack does random exit make harder?
6. Does string obfuscation protect the key?

Self-Check — Answers (1–6)

1. **MATE = Man-At-The-End**: the attacker fully possesses the device (memory, debugger, key). Crypto assumes a secure endpoint; under MATE the key is exposed **while running** → crypto alone is not enough.
2. Obfuscation does not mean unbreakable; it demands **more time/skill/tools** for the attack. If breaking 100 TL worth of content requires 10,000 TL of effort, the attacker gives up.
3. A **bogus operation** runs but its result is not used (it tires the analyst out); a **dead branch**, protected by an opaque predicate, **never runs**.
4. In flattening, the dispatcher pattern is recognisable and the state variable can be tracked. Reinforcements: **opaque predicate · encrypting the state variable · bogus states/blocks + random exit**.
5. It makes **pattern-matching and automated-script** attacks (expecting the same pattern for the same input) and symbolic execution harder.
6. **No**. String/table obfuscation makes static `strings` scanning harder, but the key is exposed **in memory while running** → whitebox/HSM is required.

Self-Check (7–12)

7. What are virtualisation's two costs?
8. Which OS protection does self-modifying code conflict with?
9. The four metrics; which ones trade off?
10. Does diversification prevent potency or scaling?
11. Which rules does KLEE break? How is resilience increased?
12. Fill in the rule template for a section.

Self-Check — Answers (7–12)

7. A large **performance** penalty (the interpreter is slow) + a **size** increase (VM + bytecode); maintenance/debugging difficulty.
8. **W^X / DEP-NX** (writable-and-executable memory is forbidden); writing to a code page needs `mprotect` / `VirtualProtect`, which is blocked or suspicious.
9. **Potency · resilience · stealth · cost**. Potency/resilience $\uparrow \rightarrow$ cost $\uparrow$ and stealth $\downarrow$ (looks abnormal). Main trade-off: **resilience $\leftrightarrow$ cost**.
10. It prevents **scaling** — it does not raise a single copy's strength; a crack does **not spread** to every copy (it breaks reuse of the attack).
11. **KLEE** is symbolic execution; it can solve opaque predicates/flattening. Rule: make predicates **resistant to symbolic execution** (input-dependent, hard to solve); path-exploding structures raise resilience.
12. Example: **Rule** [flattening] $\rightarrow$ **Goal** [hide the flow] $\rightarrow$ **How** [Tigress Flatten + opaque predicate] $\rightarrow$ **Measurement** [size +X%, speed -Y%, instructions $N \rightarrow M$]. The measurement line is required.

Glossary (1)

Term	Meaning
MATE / white box	An attacker who possesses the program
Opaque predicate	A condition whose value the programmer knows, that the tool struggles to resolve
Flattening	Moving the flow into a single switch
Bogus operation	Code that runs but does not change the result
Dead branch	A branch (protected by an opaque predicate) that never runs

Glossary (2)

Term	Meaning
Opaque boolean	A true/false value that cannot be flipped with a single byte
Virtualisation	Turning a function into custom VM bytecode
Diversification	Structurally different copies from the same source
Potency/resilience/stealth/cost	The obfuscation metrics

Sources

- **Cookbook** (Viega & Messier) Chapter 12 — anti-tampering
- **Secure programming technical guide** — native countermeasures
- Collberg & Nagra, *Surreptitious Software* — taxonomy, measurement
- Schrittwieser et al. (ACM CSUR 2016) — overview, deobfuscation
- Banescu et al. — Tigress + KLEE resilience measurement
- Obfuscator-LLVM — compiler-based

Next Week

Week 10 — Certificates and Cryptographic Methods

We said "obfuscation does not protect the key" this week; week 10 covers choosing keys correctly, their lifecycle, and protecting them with PKI. Rule (this week) → whitebox (week 11) → automation (week 14): these three weeks form one whole.

Obfuscation is not a fortress, it is a **delaying layer**: it does not give unbreakability, it raises **cost**; its strength comes from **togetherness**, **diversification**, and being **measured**.